Ex:No: 1G Date: 04.02.26	Marks Tracking and Threshold Identification Using Multidimensional Array

Problem Description:

A college maintains the **marks of students across multiple assessments and months** for academic monitoring. The marks of **students over several days, assessments, and months** are recorded. The college wants to **store this data in a multidimensional array** (Students × Days × Assessments × Months) and **identify marks below a user-defined threshold** for remedial action.

Objective:

To apply multidimensional array operations in R for storing and analyzing student marks across multiple days, assessments, and months, and to identify low-scoring records based on a user-defined threshold.

Sample Input

Enter number of students: 2
Enter number of days per assessment: 2
Enter number of assessments: 2
Enter number of months: 2
Enter minimum marks threshold: 50

Expected Output

```
--- Month 1 ---

Enter marks for Assessment 1 :
Enter marks for Student 1 for 2 days separated by space:
1: 56 64
Read 2 items
Enter marks for Student 2 for 2 days separated by space:
1: 45 52
Read 2 items

Enter marks for Assessment 2 :
Enter marks for Student 1 for 2 days separated by space:
1: 75 40
Read 2 items
Enter marks for Student 2 for 2 days separated by space:
1: 65 38
Read 2 items

--- Month 2 ---

Enter marks for Assessment 1 :
Enter marks for Student 1 for 2 days separated by space:
1: 65 85
Read 2 items
Enter marks for Student 2 for 2 days separated by space:
1: 45 75
Read 2 items

Enter marks for Assessment 2 :
Enter marks for Student 1 for 2 days separated by space:
1: 56 47
Read 2 items
Enter marks for Student 2 for 2 days separated by space:
1: 50 56
Read 2 items
```

```
Marks Array (4-D):
, , Assessment 1, Month 1

          Day 1 Day 2
Student 1    56    45
Student 2    64    52

, , Assessment 2, Month 1

          Day 1 Day 2
Student 1    75    65
Student 2    40    38

, , Assessment 1, Month 2

          Day 1 Day 2
Student 1    65    45
Student 2    85    75

, , Assessment 2, Month 2

          Day 1 Day 2
Student 1    56    50
Student 2    47    56
```

```
Low Marks Records (< 50 ) for Month 1 :

Assessment 1 :
Student 1 Day 2 : 45

Assessment 2 :
Student 2 Day 1 : 40
Student 2 Day 2 : 38

Low Marks Records (< 50 ) for Month 2 :

Assessment 1 :
Student 1 Day 2 : 45

Assessment 2 :
Student 2 Day 1 : 47
```

Program:

User input for dimensions – students, days, assessments, months and threshold

```
students <- as.integer(readline("Enter number of students: "))
days <- as.integer(readline("Enter number of days for
                                    assessment: "))
assessments <- as.integer(readline("Enter number of assessments
months <- as.integer(readline("Enter number of months: "))
threshold <- as.numeric(readline("Enter minimum marks thresho
                                    "))
```

Empty vector to store marks

```
marks_vector <- c()
```

INPUT:

Enter number of students: 1
Enter number of days per assessment: 2
Enter number of assessments: 1
Enter number of months: 1
Enter minimum marks threshold: 50
month 1
Enter marks for Assessment 1:
Enter marks for student 1 for 2 days seperated by space:
40 60

OUTPUT:

Marks Array (4-D):
Assessment 1, month 1

	Day 1	Day 2
tudent 1	40	60

Low Marks Records (<50) for month 1:
Assessment 1:
Student 1 Day 1: 40

Input marks for each month, assessment, and student

```
for(m in 1: months){
  cat("\n month", m, "\n")
  for(a in 1: assessments){
   cat("Enter marks for assessment", a, ":\n")
  for(s in 1: students) {
    input <- readline(paste("Enter marks for student ", s, "for",
                            days, "days separated by space :"));
   values <- as.numeric(strsplit(input, " ")[[1]])
   marks_vector <- c(marks_vector, values)
  }
 }
}
```

Create 4-D array

```
marks_array <- array(marks_vector,
  dim = c(students, days, assessments, months),
  dimnames = list(paste("student", 1:students),
  paste("Day", 1:days),
  paste("Assessment", 1:assessments),
  paste("month", 1:months)
  )
)
```

```
# Display the multidimensional array
cat ("\n Marks Array (4-D): \n")
for (m in 1 : months) {
  for (a in 1 : assessments) {
    cat ("\n Assessment ", a, "Month, ", m, "\n")
    print (Marks_array [, , a, m])
}
}
# Identify low marks records
for (m in 1: months) {
  cat ("\n Low marks Records (<", threshold, ") for month ", m, ":\n")
for (a in 1: assessments) {
  cat ("Assessment", a, ":\n")
  for (s in 1 : students) {
    for (d in 1 : days) {
      if (marks_array [s, d, a, m] < threshold) {
        cat (
          dimnames (marks_array) [[1]] [s],
          dimnames (marks_array) [[2]] [d],
        ": ", marks_array [s, d, a, m], "\n"
        )
}
}
}
}
}
}
```

Result:

Thus, the program successfully demonstrates the use of multidimensional arrays in R to store and analyze student marks across multiple days, assessments, and months, and to identify low-scoring records based on a user-defined threshold.